CareFlow: Clinical Pathway Process Mining - Complete Setup Guide

Intern: Yash Rahangdale | Data Associate L1 Intern | Infotact Solutions | Ref. No. ITS/DA/1393 | 10 Oct 2026 - 10 Nov 2026

0. Project kit me kya-kya hai

- data/raw/ - event_log.csv (13,664 rows) aur case_attributes.csv (1,500 cases)
- data/processed/ - event_log_clean.csv (13,583 rows, 26 columns)
- src/ - generate_data.py, clean_data.py, preprocess.py, discover_models.py, analyze_insights.py, conformance.py, ml_model.py, connect_db.py, pm4py_discovery.py, utils.py
- sql/ - schema.sql, load_data.sql, kpi_queries.sql, views.sql
- notebooks/ - 00 database, 01 understanding, 02 cleaning, 03 discovery, 04 insights, 05 ML
- dashboard/app.py - Streamlit dashboard
- outputs/ - 14 charts (PNG), JSON/CSV reports, trained model (joblib)
- docs/ - yeh guide + project report | presentation/ - PPT | run_all.py - ek click me pura pipeline

1. Start kaise kare - sahi order

Pehle project ko laptop par chalao, phir GitHub par daalo. Ghabrao mat, 6 command me ho jata hai:

```
python -m venv .venv
.venv\Scripts\activate        # Windows
source .venv/bin/activate     # Mac/Linux
pip install -r requirements.txt
python run_all.py
streamlit run dashboard/app.py
```

run_all.py 6 stages chalata hai: data generate -> clean+features -> process discovery -> insights/charts -> conformance -> ML model. 25 second me pura pipeline khatam.

2. GitHub par upload - kaun si file pehle

Rule: pehle SKELETON (README + structure + config), phir CODE, phir OUTPUT, phir DOCS. Aisa karne se har commit ka matlab samajh aata hai aur mentor ko weekly progress dikhti hai.

Order	Files	Kyun / Kab
1	README.md, .gitignore, requirements.txt, config/config.yaml	Repo ka chehra. Sabse pehla commit.
2	data/raw/*.csv, src/generate_data.py, src/utils.py, src/clean_data.py, sql/schema.sql, sql/load_data.sql	Week 1 - data ready karna.
3	src/preprocess.py, src/discover_models.py, src/conformance.py, sql/kpi_queries.sql, sql/views.sql, notebooks/	Week 2 - process mining + SQL.
4	src/analyze_insights.py, src/ml_model.py, dashboard/app.py, outputs/	Week 3 - insights, charts, dashboard, model.
5	docs/, reports/, presentation/, README (final)	Week 4 - report, PPT, documentation.

3. Git commands - step by step

```
cd CareFlow_Master
git init -b main
git config user.name  "Yash Rahangdale"
git config user.email "yashrahangdale112@gmail.com"

git add README.md .gitignore requirements.txt requirements-optional.txt config/
git commit -m "Week 1: project skeleton, README and config"

git add data/raw src/generate_data.py src/utils.py src/clean_data.py sql/schema.sql sql/load_data.sql
git commit -m "Week 1: raw clinical event log, generator and cleaning script"

git add src/preprocess.py src/discover_models.py src/conformance.py sql/kpi_queries.sql sql/views.sql notebooks/
git commit -m "Week 2: preprocessing, process discovery, conformance and SQL KPIs"

git add src/analyze_insights.py src/ml_model.py dashboard/ outputs/ src/connect_db.py src/pm4py_discovery.py run_al
git commit -m "Week 3: KPI insights, charts, Streamlit dashboard and long-stay model"

git add docs/ reports/ presentation/ README.md
git commit -m "Week 4: documentation, project report and presentation"

git remote add origin https://github.com/<your-username>/CareFlow-Clinical-Pathway-Process-Mining.git
git branch -M main
git push -u origin main
```

Baad me changes push karne ke liye sirf: git add . ; git commit -m "message" ; git push

4. GitHub web upload (git nahi chahiye)

- github.com par login -> New repository -> naam: CareFlow-Clinical-Pathway-Process-Mining -> Public -> Create.
- Add file -> Upload files -> folder me se files drag-drop karo. NOTE: browser folder drop karega to path ban jata hai; isliye src, sql, docs ko ek-ek karke upload karna.
- Neeche Commit changes likh kar save karo. Bade folder (data/) ke liye pehle git use karna aasan hai.
- Upload ke baad README automatically home page par dikhta hai.

5. Database connect kaise kare (MySQL / PostgreSQL)

Project me star schema (dim_patient, dim_activity, dim_department, dim_resource, dim_date + fact_case, fact_event) hai. 4 SQL files isi order me chalao: schema.sql -> load_data.sql -> kpi_queries.sql -> views.sql.

```
# MySQL
mysql -u root -p < sql/schema.sql
mysql -u root -p --local-infile=1 careflow < sql/load_data.sql
mysql -u root -p careflow < sql/kpi_queries.sql

# Python (SQLAlchemy) - same code teeno DB ke liye
python src/connect_db.py --engine sqlite     # demo, server ki zarurat nahi
python src/connect_db.py --engine mysql      # MySQL chal raha ho to
python src/connect_db.py --engine postgres
# URL khud ka dena ho to:  export CAREFLOW_DB_URL="mysql+pymysql://user:pass@localhost:3306/careflow"
```

Zaroori baat: MySQL me TIMESTAMP reserved word hai, isliye column ka naam event_timestamp rakha gaya hai. event_index ko ROW_NUMBER() se dobara banao - yehi case ke andar event ka order hai jo process mining ke liye must hai.

6. Data cleaning - kya-kya karna hai

#	Step	Result
1	Column names trim + lower	8 clean columns
2	timestamp -> datetime, numbers -> numeric	type errors khatam
3	case_id / activity / timestamp blank rows drop	0 rows (log saaf tha)

#	Step	Result
4	Duplicate events hataye (same case+activity+timestamp)	81 duplicates removed
5	Blank resource -> UNKNOWN, department -> Unassigned	203 events bache, delete nahi kiye
6	Negative wait/cost -> 0, service_minutes median se bharo	outliers handle
7	event_index, gap_minutes, is_rework banao	process mining ready
8	3 se chhote / 200 se bade case drop	0 cases

Final: 13,664 -> 13,583 events, 1,500 cases, 15 activities. Audit file: outputs/reports/cleaning_audit.json

7. Dashboard - Streamlit (recommended) + Power BI

```
streamlit run dashboard/app.py     # http://localhost:8501
```

- Streamlit tabs: Overview KPI, Process (DFG + variants), Bottlenecks, Resources, Conformance, ML prediction.
- Power BI: MySQL server se 'Get Data' -> v_powerbi_export view import karo. Ya outputs/reports/*.csv import kar lo.
- Power BI pages: KPI cards (cases, avg LOS, long-stay %), ribbon/bar for bottlenecks, heatmap weekday x hour, cost by department, LOS histogram.
- Report ke saath screenshot lena - docs/screenshots/ me rakho.

8. Real hospital log use karna ho to

Public real log: 4TU.ResearchData par 'Sepsis Cases - Event Log' (Mannhardt, ~1,050 cases / 15,214 events, 16 activities). Uska schema alag hota hai, isliye rename mapping chahiye: case:concept:name -> case_id, concept:name -> activity, time:timestamp -> timestamp, org:resource -> resource. Uske baad:

```
python src/clean_data.py          # cleaning dobara
python run_all.py --skip-data     # pipeline real data par
```

Synthetic log se fitness 1.0 aata hai (generator fixed templates follow karta hai). Real log par 0.7-0.95 aur long-stay AUC 0.75-0.90 expect karo - report me ye baat likhna zaroori hai.

9. Deliverables checklist (assignment letter ke hisaab se

Deliverable	File
Source Code	src/, dashboard/, sql/, notebooks/, run_all.py
Project Documentation	docs/CareFlow_Setup_Guide_Hinglish.pdf + docs/documentation.html
Project Report	reports/CareFlow_Project_Report.pdf
Presentation (PPT)	presentation/CareFlow_Presentation.pptx
Model Evaluation	outputs/reports/model_metrics.json + model_evaluation PDF
Demonstration	streamlit run dashboard/app.py

10. Common error aur fix

- 'No module named pm4py' -> optional hai, chalta rehta hai; real miners ke liye pip install -r requirements-optional.txt + Graphviz.
- 'Access denied' MySQL me -> LOAD DATA LOCAL INFILE ke liye --local-infile=1 lagao.

- Streamlit blank -> folder ke root se chalao (jahan .streamlit/dashboard hai), path relative hote hain.
- Excel/CSV me timestamp text ban jaye -> pd.read_csv(parse_dates=['timestamp']).
- Graphviz error on inductive miner image -> sudo apt-get install -y graphviz (Windows me PATH set karo).